pins 18 and 19 of IC6 along with two 33pF capacitors (C2 and C3). This circuitry provides clock signal to the MCU for its internal operations. Capacitor C1 in parallel with resistor R16 and push button S1 are connected to RST pin 9 to provide manual reset to the MCU.

When any sensor detects motion, data is transmitted through RF Tx. This is received and demodulated by RF Rx module. This data is sent to RF decoder HT12D (IC5), which detects valid address indicated through the blinking of LED5. Then, the decoder latches the data, which is made available across data pins D8 through D11 of IC5. When valid address is received, VT pin of IC5 goes high and the MCU gets an interrupt signal. It gets bit pattern on port P1. As per data bits, any one pin (out of four) will be low and rest will be high. Based on which pin is low, the MCU identifies the sensor that detects motion and indicates the direction of the intrusion. This is listed in Table II.

As per Table II, the MCU immediately decides and displays the message "intruder detected from xxxx side" on LCD1. It also enables both multivibrators by applying high input on RST pin.

The multivibrator (IC7) that flashes LED6 has a flashing rate of 2Hz. Component values for R19, R20 and C7 are chosen in a way that output frequency is around 2Hz.

Another multivibrator (IC8) generates 1kHz audio tone as siren through speaker (LS1). Component values for R21, R22 and C5 are chosen in a way that output frequency is around 1kHz.

When sensor PS1 (or any sensor) detects motion, its output remains high for around five to ten seconds because of a slower response. The siren sounds and LED6 continues flashing for about ten seconds. The message on LCD1 shows the side where motion is detected. After sensor output becomes stable (low), both the siren and LED6 turn off.

TABLE II
MOTION DETECTED DIRECTION

Data bit pattern of HT12D				Port P1 pins status				Location
D11	D10	D9	D8	P1.3	P1.2	P1.1	P1.0	
1	1	1	0	High	High	High	Low	Motion detected from front side
1	1	0	1	High	High	Low	High	Motion detected from back side
1	0	1	1	High	Low	High	High	Motion detected from left side
0	1	1	1	Low	High	High	High	Motion detected from right side

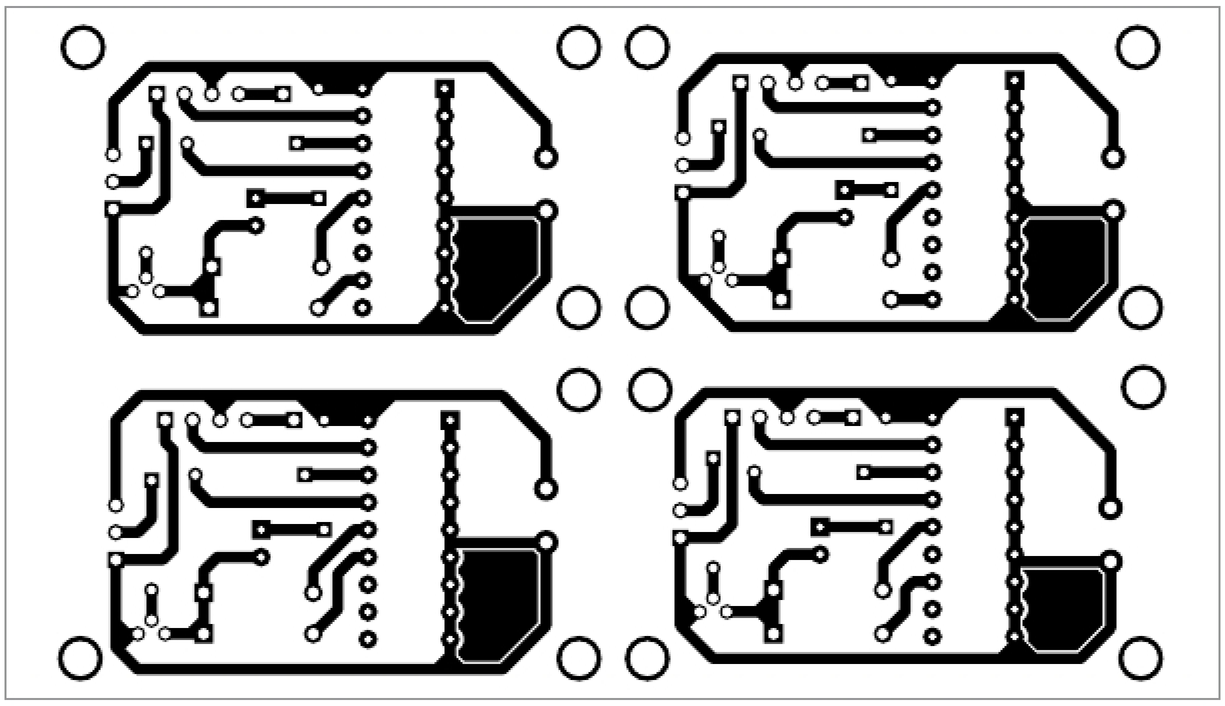

Fig. 5: PCB layout of transmitter circuits

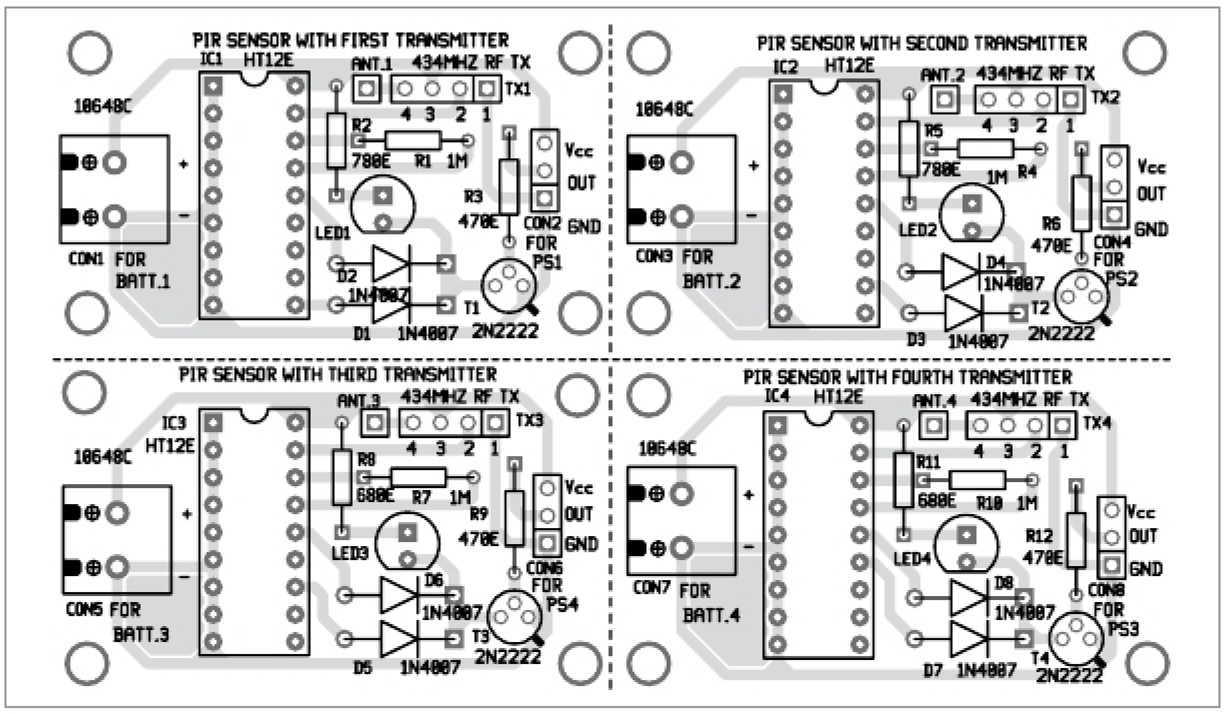

Fig. 6: Components layout for transmitter PCBs

Software program

Complete system operation is carried out based on the program embedded into the internal flash of AT89S52. The program performs the following tasks:

1. Decides where motion is detected
2. Displays warning and other messages on LCD1
3. Displays side where motion is detected
4. Flashes LED6 and turns on siren when motion is detected

The program is written in C language for 8051 family of controllers. It is compiled in Keil IDE software. The complete program is a combination of many different functions, including:

lcd_delay() generates delay for the LCD to get ready to accept a new command or data byte

lcd_send_cmd() sends command byte to LCD

lcd_send_data() sends character to be displayed on LCD

lcd_display_string() displays various strings and messages on LCD

lcd_init() configures and initialises LCD